



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab Programs

SET 16- Customer Demographics Analysis

16A. Line Chart

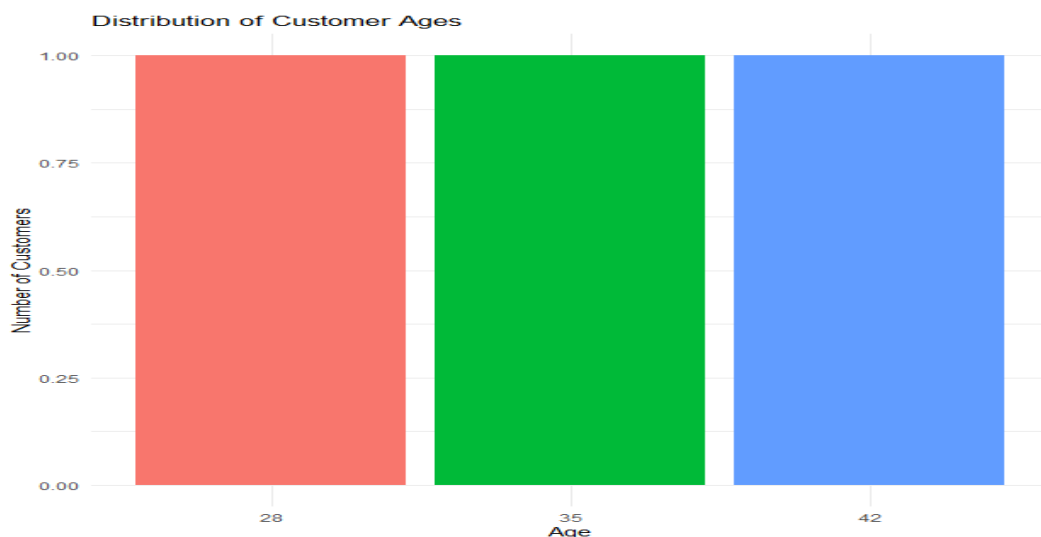
Program:

```
library(ggplot2)

customer_data <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

ggplot(customer_data, aes(x = factor(Age), fill = factor(Age))) +
  geom_bar() +
  labs(
    title = "Distribution of Customer Ages",
    x = "Age",
    y = "Number of Customers"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Output:



16B. Bar Chart

Program:

```
library(ggplot2)

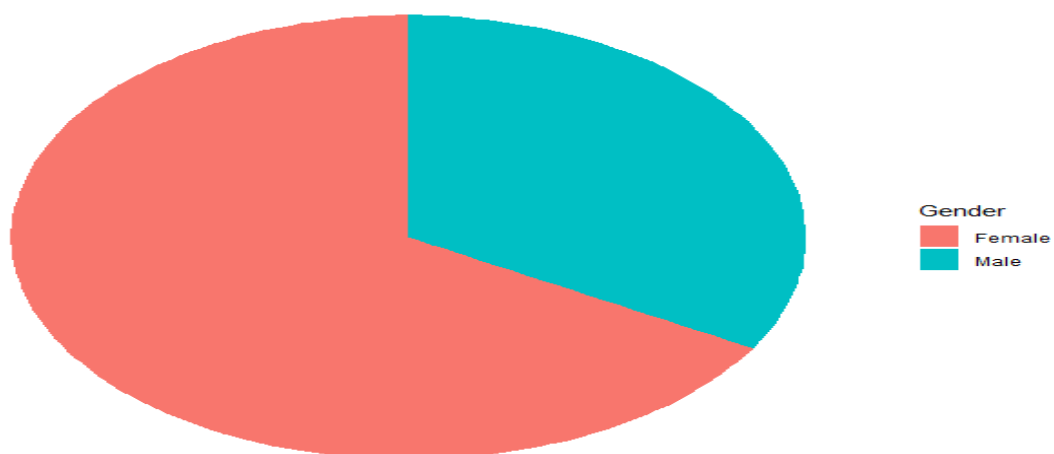
customer_data <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)

gender_data <- as.data.frame(table(customer_data$Gender))
colnames(gender_data) <- c("Gender","Count")

ggplot(gender_data, aes(x = "", y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Distribution of Customers by Gender") +
  theme_void()
```

Output:

Distribution of Customers by Gender

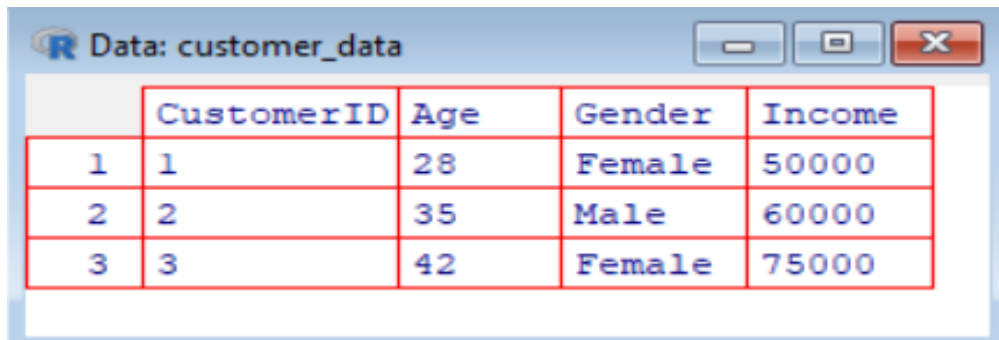


16C. Table

Program:

```
View(customer_data)
```

Output:



	CustomerID	Age	Gender	Income
1	1	28	Female	50000
2	2	35	Male	60000
3	3	42	Female	75000

16D. Tableau Dashboard

Program:

CustomerID, Age, Gender, Income

1, 28, Female, 50000

2, 35, Male, 60000

3, 42, Female, 75000

Output:

CustomerID	Age	Gender	Income	
1	28	Female	50000	
2	35	Male	60000	
3	42	Female	75000	

SET 17- Employee Performance Analysis

17A. Line Chart

Program:

```
library(ggplot2)
```

```
employee_data <- data.frame(
```

```
  EmployeeID = c(1,2,3),
```

```
  Department = c("Sales", "HR", "Marketing"),
```

```
  YearsOfService = c(5,3,7),
```

```
  PerformanceScore = c(85,92,78)
```

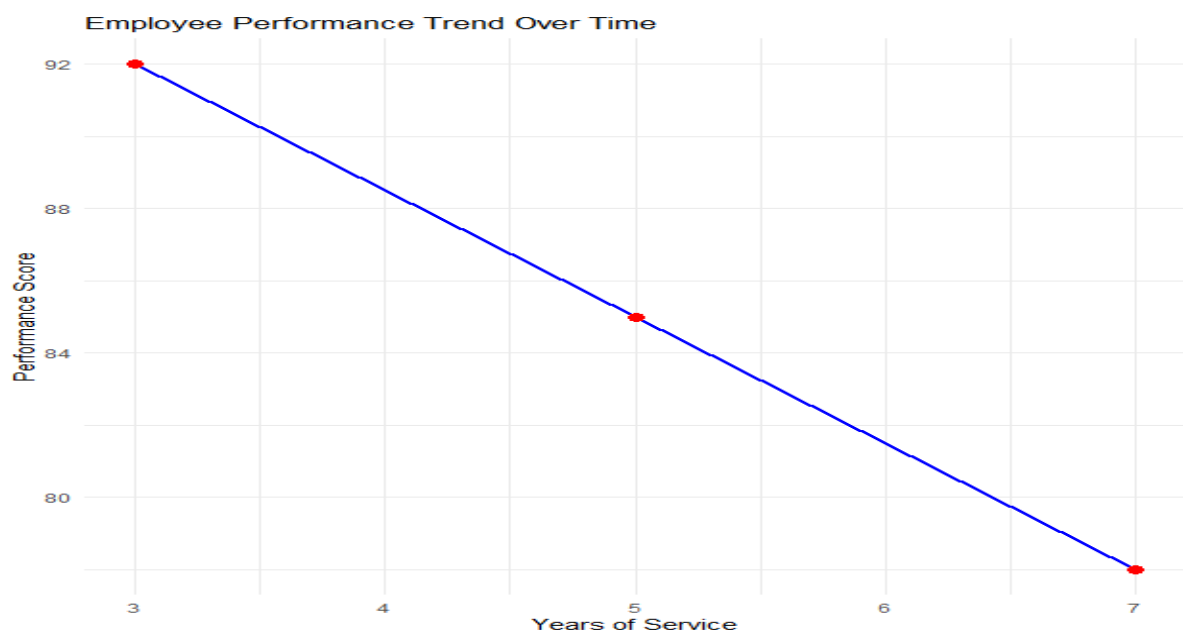
```
)
```

```
ggplot(employee_data,
```

```
  aes(x = YearsOfService, y = PerformanceScore, group = 1)) +
```

```
geom_line(color = "blue", linewidth = 1) +
geom_point(size = 3, color = "red") +
labs(
  title = "Employee Performance Trend Over Time",
  x = "Years of Service",
  y = "Performance Score"
) +
theme_minimal()
```

Output:



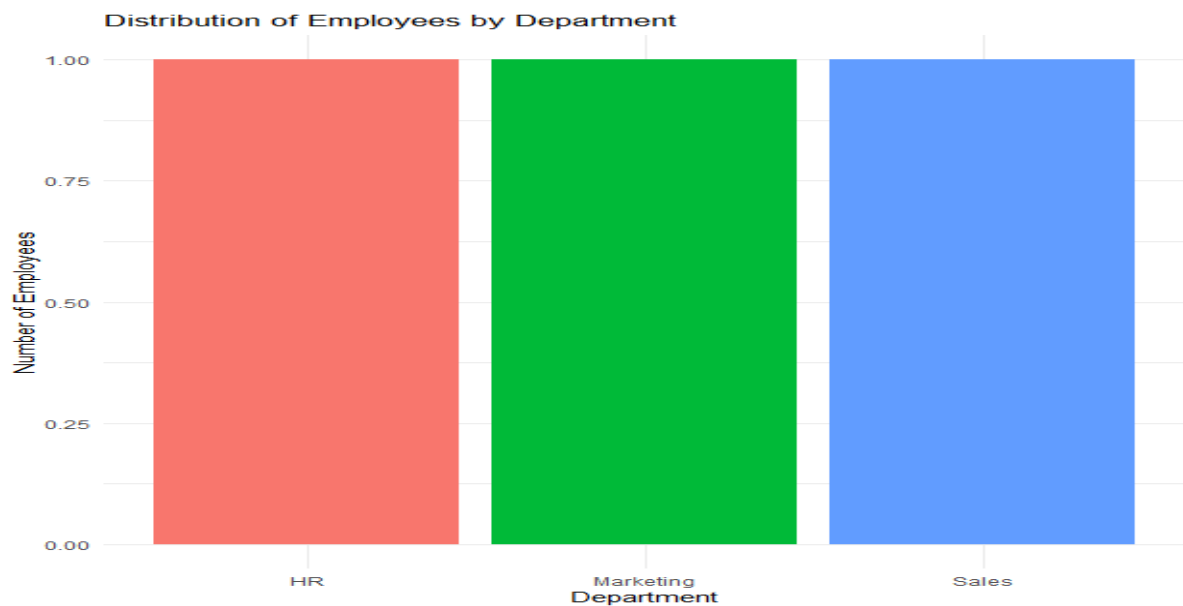
17B. Bar Chart

Program:

```
library(ggplot2)
ggplot(employee_data,
  aes(x = Department, fill = Department)) +
geom_bar() +
labs(
  title = "Distribution of Employees by Department",
  x = "Department",
  y = "Number of Employees"
) +
```

```
theme_minimal() +
theme(legend.position = "none")
```

Output:



17C. Table

Program:

```
View(employee_data)
```

Output:

Data: employee_data				
	EmployeeID	Department	YearsOfService	PerformanceScore
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

17D. Tableau Dashboard

Program:

```
write.csv(employee_data,
          "Employee_Performance.csv",
          row.names = FALSE)
```

Output:

A	B	C	D	E	
EmployeeID	Department	YearsOfService	PerformanceScore		
1	Sales	5	85		
2	HR	3	92		
3	Marketing	7	78		

SET 18- Product Inventory Management

18A. Bar Chart

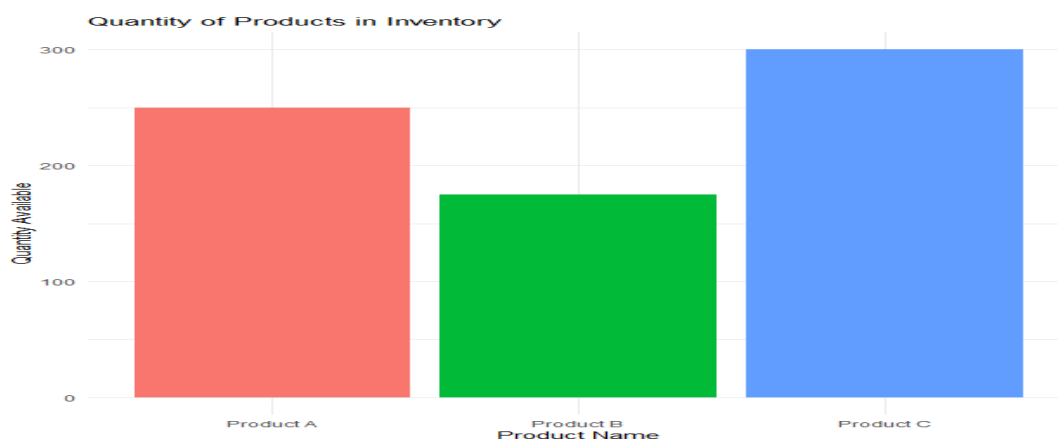
Program:

```
library(ggplot2)

inventory_data <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  QuantityAvailable = c(250,175,300),
  Price = c(20,15,18)
)

ggplot(inventory_data,
  aes(x = ProductName, y = QuantityAvailable, fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Quantity of Products in Inventory",
    x = "Product Name",
    y = "Quantity Available"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Output:



18B. Stacked Bar Chart

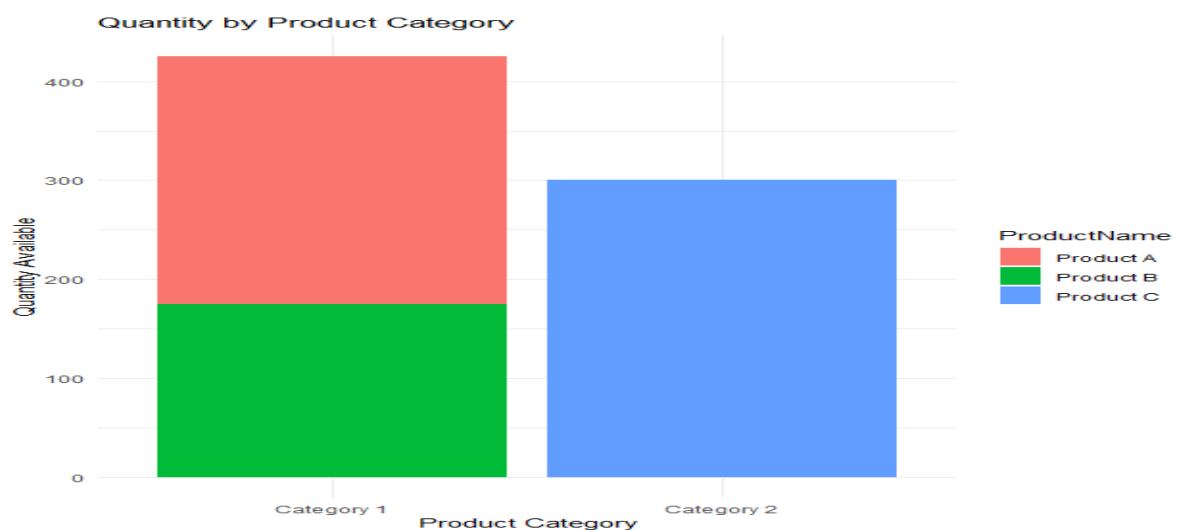
Program:

```
library(ggplot2)

inventory_data$Category <- c("Category 1","Category 1","Category 2")

ggplot(inventory_data,
       aes(x = Category, y = QuantityAvailable, fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Quantity by Product Category",
    x = "Product Category",
    y = "Quantity Available"
  ) +
  theme_minimal()
```

Output:



18C. Bar Chart

Program:

```
View(inventory_data)
```

Output:

Data: inventory_data					
	ProductID	ProductName	QuantityAvailable	Price	Category
1	1	Product A	250	20	Category 1
2	2	Product B	175	15	Category 1
3	3	Product C	300	18	Category 2

18D. Tableau Dashboard

Program:

```
write.csv(  
  inventory_data,  
  "Product_Inventory.csv",  
  row.names = FALSE  
)
```

Output:

ProductID	ProductNa	QuantityAv	Price	Category
1	Product A	250	20	Category 1
2	Product B	175	15	Category 1
3	Product C	300	18	Category 2

SET 19- Survey Responses Analysis

19A. Grouped Bar Chart

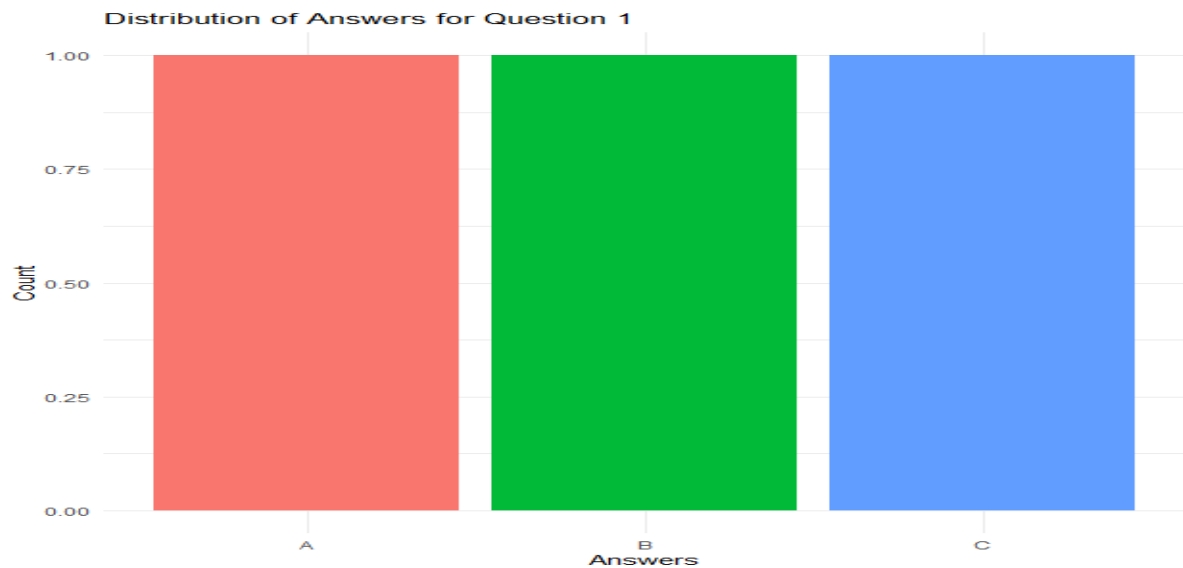
Program:

```
library(ggplot2)  
survey_data <- data.frame(  
  SurveyID = c(1,2,3),  
  Question1 = c("A","B","C"),  
  Question2 = c("B","A","A"),  
  Question3 = c("C","D","B")  
)  
  
q1_data <- as.data.frame(table(survey_data$Question1))  
colnames(q1_data) <- c("Answer","Count")  
ggplot(q1_data, aes(x = Answer, y = Count, fill = Answer)) +  
  geom_bar(stat = "identity") +  
  labs(  
    title = "Distribution of Answers for Question 1",  
    x = "Answers",  
    y = "Count"
```



```
) +  
theme_minimal() +  
theme(legend.position = "none")
```

Output:



19B. Stacked Bar Chart

Program:

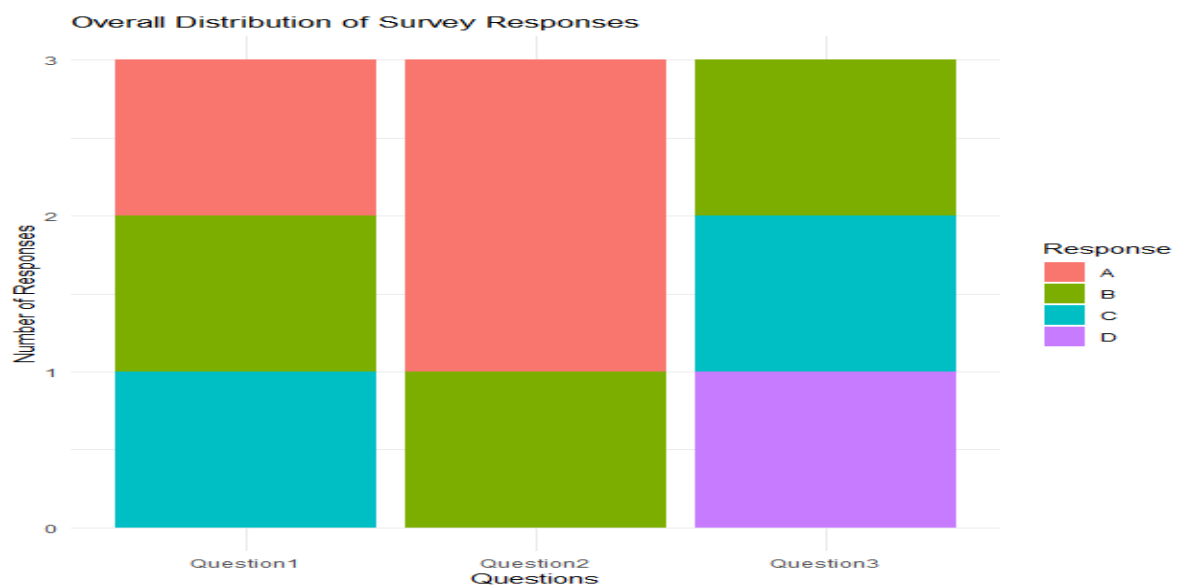
```
library(ggplot2)  
library(tidyr)  
survey_long <- pivot_longer(  
  survey_data,  
  cols = starts_with("Question"),  
  names_to = "Question",  
  values_to = "Response"  
)  
ggplot(survey_long,  
  aes(x = Question, fill = Response)) +  
  geom_bar() +  
  labs(  
    title = "Overall Distribution of Survey Responses",  
    x = "Questions",
```

```

    y = "Number of Responses"
  ) +
  theme_minimal()library(ggplot2)
library(tidyr)
survey_long <- pivot_longer(
  survey_data,
  cols = starts_with("Question"),
  names_to = "Question",
  values_to = "Response"
)
ggplot(survey_long,
  aes(x = Question, fill = Response)) +
  geom_bar() +
  labs(
    title = "Overall Distribution of Survey Responses",
    x = "Questions",
    y = "Number of Responses"
  ) +
  theme_minimal()

```

Output:

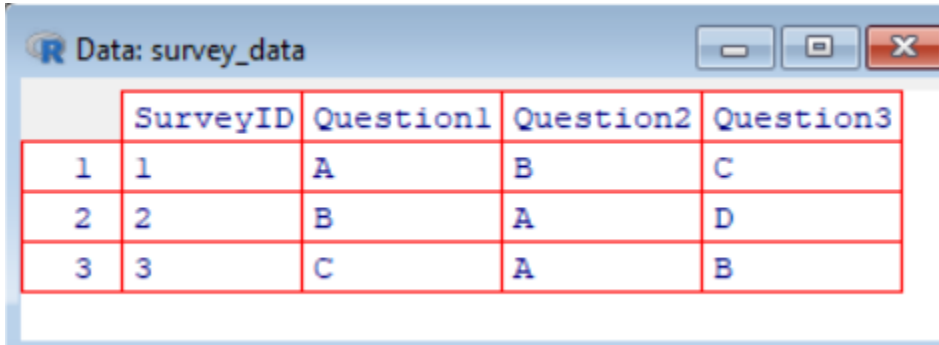


19C. Table

Program:

```
View(survey_data)
```

Output:



	SurveyID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

19D. Tableau Dashboard

Program:

```
write.csv(  
  survey_data,  
  "Survey_Responses.csv",  
  row.names = FALSE  
)
```

Output:

SurveyID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

SET 20- Stock Analysis

20A. Line Chart

Program:

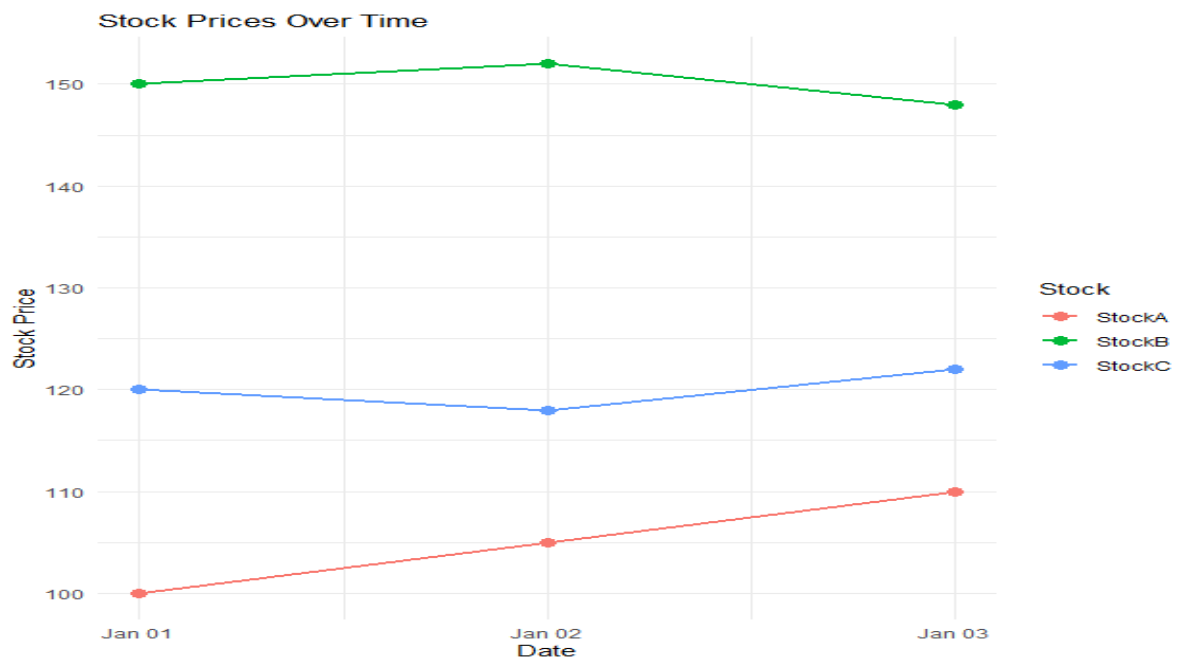
```
library(ggplot2)  
library(tidyr)  
stock_data <- data.frame(  
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03")),  
  StockA = c(100, 105, 110),  
  StockB = c(150, 152, 148),
```

```

StockC = c(120,118,122)
)
stock_long <- pivot_longer(
  stock_data,
  cols = c(StockA, StockB, StockC),
  names_to = "Stock",
  values_to = "Price"
)
ggplot(stock_long,
  aes(x = Date, y = Price, color = Stock, group = Stock)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(
    title = "Stock Prices Over Time",
    x = "Date",
    y = "Stock Price"
  ) +
  theme_minimal()

```

Output:



20B. Bar Chart

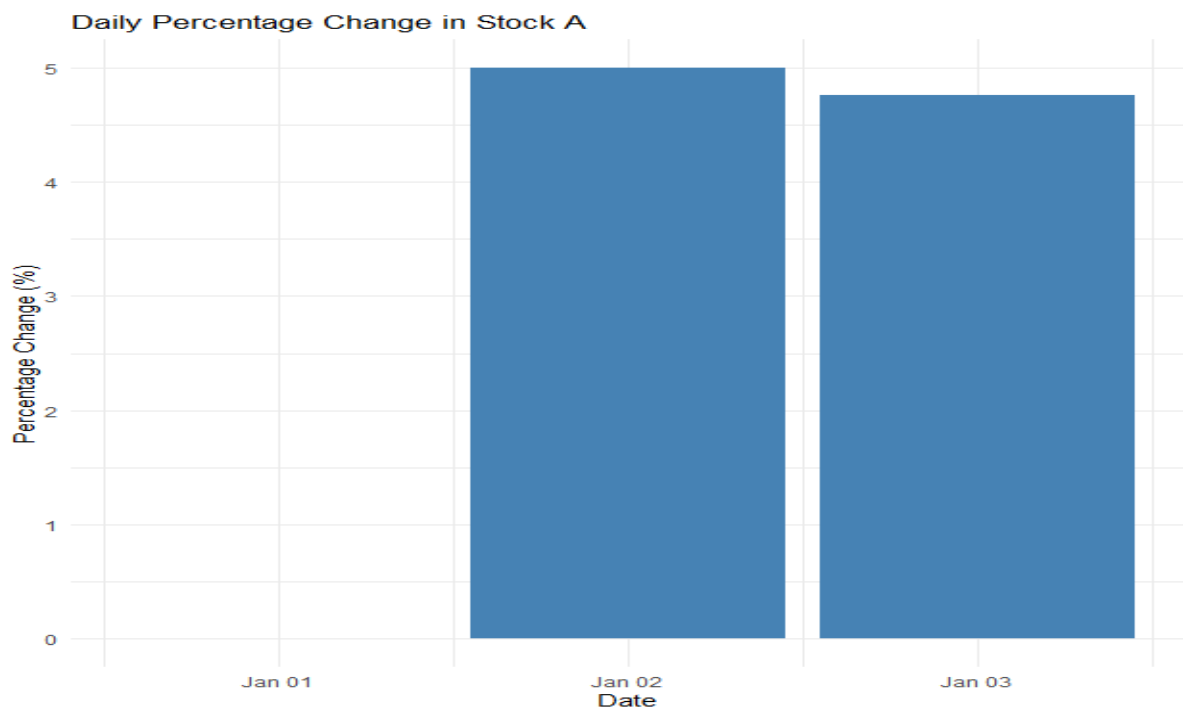
Program:

```
library(ggplot2)

stock_data$PercentageChange <- c(
  0,
  (105 - 100) / 100 * 100,
  (110 - 105) / 105 * 100
)

ggplot(stock_data,
       aes(x = Date, y = PercentageChange)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(
    title = "Daily Percentage Change in Stock A",
    x = "Date",
    y = "Percentage Change (%)"
  ) +
  theme_minimal()
```

Output:

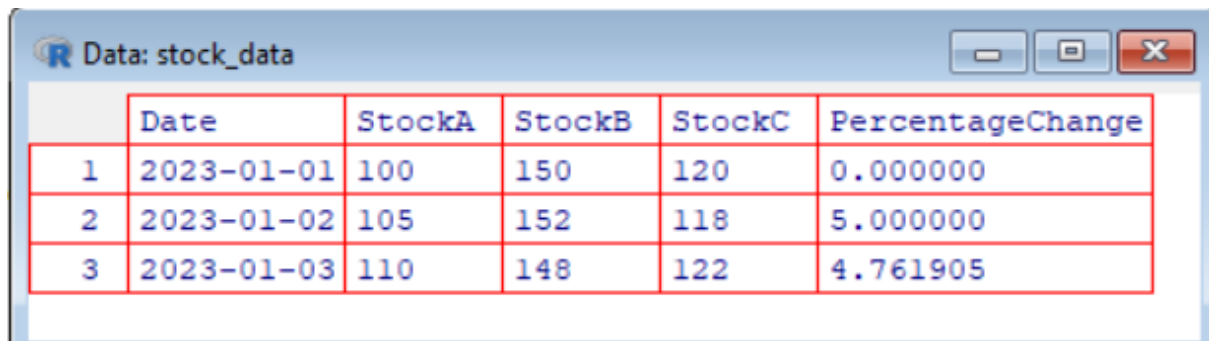


20C. Table

Program:

```
View(stock_data)
```

Output:

A screenshot of an R Data Viewer window titled "Data: stock_data". The window displays a table with 6 columns: Date, StockA, StockB, StockC, and PercentageChange. The table contains 3 rows of data. The first row has Date "2023-01-01", StockA "100", StockB "150", StockC "120", and PercentageChange "0.000000". The second row has Date "2023-01-02", StockA "105", StockB "152", StockC "118", and PercentageChange "5.000000". The third row has Date "2023-01-03", StockA "110", StockB "148", StockC "122", and PercentageChange "4.761905".

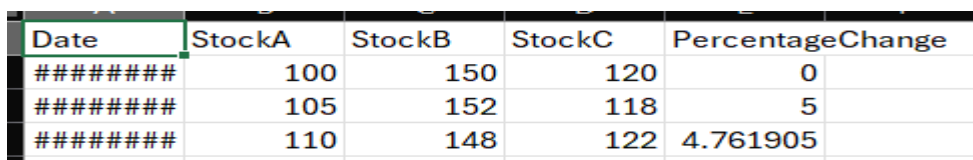
	Date	StockA	StockB	StockC	PercentageChange
1	2023-01-01	100	150	120	0.000000
2	2023-01-02	105	152	118	5.000000
3	2023-01-03	110	148	122	4.761905

20D. Tableau Dashboard

Program:

```
write.csv(  
  stock_data,  
  "Stock_Prices.csv",  
  row.names = FALSE  
)
```

Output:

A screenshot of a Tableau output table. The table has 6 columns: Date, StockA, StockB, StockC, and PercentageChange. The first column is labeled "Date" and the other five are labeled "StockA", "StockB", "StockC", and "PercentageChange". The table contains 3 rows of data. The first row has Date "#####", StockA "100", StockB "150", StockC "120", and PercentageChange "0". The second row has Date "#####", StockA "105", StockB "152", StockC "118", and PercentageChange "5". The third row has Date "#####", StockA "110", StockB "148", StockC "122", and PercentageChange "4.761905".

Date	StockA	StockB	StockC	PercentageChange
#####	100	150	120	0
#####	105	152	118	5
#####	110	148	122	4.761905